Ayush Pathak

Senior iOS Engineer | Swift, SwiftUI | Architecture, Test Automation, CI/CD | AI-Assisted Development
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Professional Summary

Senior iOS Engineer with 8+ years of experience shipping and operating production iOS apps and SDKs in Swift, SwiftUI and UIKit. Technical lead of an internal Swift package running inside a consumer app with 1M+ active users that supports a B2B business, with end-to-end ownership from design and release to production monitoring. Experienced in architecture and technology selection, test automation (unit-test coverage raised from ~25% to ~70%), mobile CI/CD, and improving app stability and quality. Work with product managers, designers, backend engineers and QA to clarify requirements through technical spikes, and mentor engineers through code and architecture reviews. Daily user of GitHub Copilot and ChatGPT Codex, and creator of custom AI agent skills adopted across the team.

Professional Experience

Numerator Inc.

Dec 2022 – Present (3+ years) | Senior iOS Engineer I (iOS Technical Lead)

- **SDK & package design:** Lead engineer of the Link Swift package, an internal SDK for market data collection inside a consumer app with 1M+ active users that supports a B2B business. Built it as a self-contained Swift Package Manager module, first as sole engineer and now with one co-owner, and am extending it for reuse in 4–5 internal apps (est.).
- **Architecture & technology selection:** Make architecture decisions jointly with a fellow senior engineer (MVVM, async/await, UIKit-to-SwiftUI migration using generics). Designed major local database, analytics and core modules, and contributed to the decision to move from Realm to SwiftData after Realm moved to community-only maintenance.
- **Performance & memory:** Diagnosed a leak in which 3–4 concurrent WebViews were never deallocated, using Instruments and the Memory Graph Debugger. Fixed it, cutting ~200 MB memory spikes (roughly 80%) back to the normal 30–40 MB range.
- **Production reliability:** Detected a crash-rate spike in Firebase Crashlytics, halted the release, reproduced the issue, wrote a regression test, fixed and verified it, then shipped and monitored a new release.
- **Test automation & QA:** Raised unit-test coverage from ~25% to ~70% in one year by requiring tests for every feature, bug fix and touched legacy code (TDD for bugs; XCTest, XCUITest, Swift Testing). Run a two-layer process with QA: CI-enforced unit tests on every PR, plus AI-generated edge-case scenarios that I curate for QA regression testing.
- **CI/CD & automation:** Maintain GitHub Actions pipelines that run tests, publish test summaries and report SwiftLint issues on PRs. Built a pipeline that automatically builds, code-signs and publishes an installable build link on a GitHub-hosted page for every PR that adopts a new package version.
- **AI-assisted engineering:** Use GitHub Copilot and ChatGPT Codex for test generation, refactoring, debugging, documentation and planning (est. ~70% of code written with AI assistance). Built internal developer tooling as custom Markdown AI agent skills for code review, project conventions and reusable workflows, used by the whole team. Familiar with MCP.
- **Collaboration & mentoring:** Work with product managers, designers and backend engineers on requirements, technical spikes, risks (backward compatibility, data integrity) and API design, including request/response models and security headers. Mentor engineers, conduct technical interviews, take part in code and architecture reviews, join two-week Jira sprint planning, and present hack-day prototypes to the company.

Appentus Technologies Pvt Ltd

May 2018 – Nov 2022 (4.6 years) | Sr. iOS Engineer

- **Leadership:** Led and mentored iOS developers (teams of up to 4) and oversaw project execution, code reviews and task planning with GitHub and Jira.
- **Architecture & data:** Built apps with MVC and MVVM, SOLID principles and design patterns (singleton, repository, decorator). Used Realm and Core Data, and web services over REST/JSON, XML, Protobuf and gRPC.
- **CI/CD & quality:** Implemented CI/CD with Fastlane, CircleCI and Xcode Cloud for automated builds, testing and deployment. Wrote unit tests and enforced code quality with SwiftLint.
- **Features & integrations:** Delivered in-app purchases, localization, Firebase, Google Maps and socket integrations, using GCD, operation queues and Combine for asynchronous work.

Projects

Link Swift Package (Numerator, internal SDK)
Jan 2023 – Present | Lead iOS Developer

- Internal SDK that takes control from the host app, collects data through active and background WebViews, stores it locally (Realm, Keychain) and hands results back; uses Firebase Remote Config, Statsig and Crashlytics.

Teesas – Education
Feb 2022 | Lead iOS Developer | View on App Store

- Worked with backend, UI and UX teams to plan delivery. Built an eBook and course marketplace with a smooth purchase flow (in-app purchases, Apple Pay), video player, PDF viewer and downloads. Implemented URLSession web services, SSL pinning, Firebase push notifications and Core Data to reduce load time.

Hoofzy – Pet Care and Social App
Aug 2022 | Lead iOS Developer in a team of two | View on App Store

- Built a pixel-perfect iPhone UI with HTTP/2 web services via gRPC and Protobuf. Used Realm to improve performance, added unit tests across modules to increase coverage, and integrated OneSignal, Crashlytics, in-app purchases and SwiftLint.

Education

Bachelor of Technology in Computer Engineering
2015 – 2019

Achievements & Awards

- Numerator Star Performer Award – Q3 2026
- Numerator Star Performer Award – Q4 2025
- Numerator Spot Award – 2024, for outstanding performance
- Attended try! Swift Tokyo 2025

Technical Skills

Languages: Swift 5
iOS Frameworks: UIKit, SwiftUI, Combine, Swift Concurrency (async/await), GCD, Operation Queues, Core Data, SwiftData, Realm, Keychain, WebView, In-App Purchases, Apple Pay, Localization
Architecture & Design: MVVM, MVC, SOLID, Design Patterns, Modular Design, SDK / Reusable Package Design, Swift Package Manager, CocoaPods; TCA and Clean Architecture (conceptual familiarity)
Testing & Quality: XCTest, XCUITest, Swift Testing, Unit Testing, Test-Driven Development, Test Automation, QA Collaboration, SwiftLint, Code Review, Instruments, Memory Graph Debugger
CI/CD & Tools: GitHub Actions, Fastlane, CircleCI, Xcode Cloud, Code Signing, Xcode, Git, GitHub, Jira, Figma
APIs & Networking: REST, JSON, XML, gRPC, Protocol Buffers, URLSession, WebSockets, SSL Pinning, API Security, API Design with Backend Teams; GraphQL / Apollo iOS (familiarity)
Monitoring & Services: Firebase (Crashlytics, Remote Config, Push Notifications), Statsig, OneSignal, Google Maps
AI-Assisted Development: GitHub Copilot, ChatGPT Codex, Custom Agent Skills (Markdown), AI Agents, Internal Developer Tooling, AI Code Review, AI Test Generation, MCP (familiarity)
Methodologies: Agile / Scrum, Sprint Planning, Story Point Estimation, Technical Spikes, Release Management, Technical Mentoring, Technical Interviewing, Cross-functional Collaboration
Spoken Languages: English (Fluent), Hindi (Native), Japanese (JLPT N5, currently studying)